

# Mohammad Naimul Islam Suvon

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## Research Interests

Multimodal AI, Cardiac Imaging, Low-cost Clinical AI, Deployment-centric AI, Open-source Software

## Selected Appointments

- AI Research Engineer – Centre for Machine Intelligence (CMI), the University of Sheffield 09.2023 – Present
- Research Assistant - Machine Learning for Medical Image Analysis, the University of Sheffield 04.2023 – 08.2023
- Research & Course Support – Turing Network Development Award, the University of Sheffield 09.2022 – 03.2023

## Education

- PhD in Computer Science (Part-time) – University of Sheffield, UK 02.2024 – Present
- MSc in Computer Science with Speech and Language Processing – University of Sheffield, UK 09.2021 – 09.2022
- BSc in Computer Science and Engineering – North South University, Bangladesh 01.2017 – 08.2020

## Publications (h-index: 8, i10-index: 6, Total Citations: 159)

### Journal Papers

- [J1] X. Liu, J. Zhang, S. Zhou, ..., **M. N. I. Suvon**, *et al.*, “Towards deployment-centric multimodal AI beyond vision and language,” ***Nature Machine Intelligence (IF: 23.9)***, 7, pp. 1612–1624, 2025. DOI: 10.1038/s42256-025-01116-5.
- [J2] P. C. Tripathi, S. Tabakhi, **M. N. I. Suvon**, L. Schöb, S. Alabed, A. J. Swift, S. Zhou and H. Lu (2025), “Interpretable multimodal learning for cardiovascular hemodynamics assessment,” ***IEEE Transactions on Medical Imaging (IF: 9.8)***. arXiv: 2404.04718, Accepted with minor revision.
- [J3] **M. N. I. Suvon**, S. Zhou, P. C. Tripathi, W. Fan, S. Alabed, B. Khanal, V. Osmani, A. J. Swift, C. (Cherise) Chen and H. Lu, “Multimodal Latent Fusion of ECG Leads for Early Assessment of Pulmonary Hypertension,” ***IEEE Journal of Biomedical and Health Informatics (IF: 6.7)***. arXiv: 2503.13470, Under major revision.
- [J4] S. Tabakhi, **M. N. I. Suvon**, P. Ahadian and H. Lu, “Multimodal Learning for Multi-omics: A Survey,” *World Scientific Annual Review of Artificial Intelligence*, 1, 2023, DOI: 10.1142/S2811032322500047.
- [J5] **M. N. I. Suvon**, M. Alam, S. C. Siam, M. Ferdous and R. Khan, “Masters and Doctor of Philosophy admission prediction of Bangladeshi students into different classes of universities,” *IAES International Journal of Artificial Intelligence*, 11(4), pp.1545-1553, 2022. DOI: 10.11591/ijai.v11.i4.pp1545-1553.
- [J6] F. Noor, M. R. Tanjim, M. J. Rahim, **M. N. I. Suvon**, F. K. Porna, S. Ahmed, M. A. A. Kaiousm and R. Rahman., “Application of fuzzy logic on CT-scan images of COVID-19 patients,” *International Journal of Intelligent Information and Database Systems*, 14(4), pp.333-348, 2021. DOI: 10.1504/IJIDS.2021.118561.

### Conference Papers

- [C1] **M. N. I. Suvon**, P. C. Tripathi, W. Fan, S. Zhou, X. Liu, S. Alabed, V. Osmani, A. Swift, C. Chen, & H. Lu, “Multimodal Variational Autoencoder for Low-cost Cardiac Hemodynamics Instability Detection”, in *the 27th Int. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI, CORE A)*, pp. 296–306, Oct. 2024. DOI: 10.1007/978-3-031-72378-0.
- [C2] P. C. Tripathi, **M. N. I. Suvon**, S. Zhou, S. Alabed, A. J. Swift and H. Lu., “Tensor-based Multimodal Learning for Prediction of Pulmonary Arterial Wedge Pressure from Cardiac MRI”, in *26th Int. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI, CORE A)*, pp. 206–215, Oct. 2023. DOI: 10.1007/978-3-031-43990-2.
- [C3] **M. N. I. Suvon**, P. C. Tripathi, S. Alabed, A. J. Swift and H. Lu., “Multimodal Learning for Predicting Mortality in Patients with Pulmonary Arterial Hypertension,” in *Proc. of the 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 2704–2710, 2022 DOI: 10.1109/BIBM55620.2022.9995597.
- [C4] M. M. Alam, **M. N. I. Suvon**, and R. Khan., “Social Distance Measurement and Face Mask Detection Using Deep Learning Models,” in *Proc. of the 2021 International Conference on Intelligent Emerging Methods of Artificial Intelligence & Cloud Computing*, pp. 540–549, 2021. DOI: 10.1007/978-3-030-92905-3\_66.
- [C5] S. C. Siam, A. Faisal, N. Mahrab, A. B. Haque and **M. N. I. Suvon.**, “Automated Student Review System with Computer Vision and Convolutional Neural Network”, in *Proc. of the 2021 International Conference on Computing, Communication, and Intelligent Systems*, pp.493-497, 2021. DOI: 10.1109/ICCCIS51004.2021.9397164.

- [C6] A. Iqbal, M. F. Ahmed, **M. N. I. Suvon**, S. D. Shuvo, and A. Fahmin., “Towards Efficient Segmentation and Classification of White Blood Cell Cancer Using Deep Learning”, in *Proc. of the 2021 Emerging Technology in Computing, Communication and Electronics*, pp. 1-6, 2021. DOI: 10.1109/ETCCE54784.2021.9689839.
- [C7] F. Mashroor, I. F. Ishrak, S. M. Alvee, A. Jahan, **M. N. I Suvon** and S. Siddique., “Rice Paddy Disease Detection and Disease Affected Area Segmentation Using Convolutional Neural Networks,” in *Proc. of the 2021 IEEE Region 10 Conference (TENCON)*, pp. 891-896, 2021. DOI: 10.1109/TENCON54134.2021.9707192.
- [C8] **M. N. I. Suvon**, M. Ferdous and R. Khan., “Real Time Bangla Number Plate Recognition using Computer Vision and Convolutional Neural Network,” in *Proc. of the 2020 IEEE 2nd International Conference on Artificial Intelligence in Engineering and Technology*, pp. 1–6, 2020. DOI: 10.1109/IICAET49801.2020.9257843.

## Preprints & Abstracts

- [P1] W. Fan, **M. N. I. Suvon**, S. Zhou, X. Liu, S. Alabed, V. Osmani, A. Swift, C. Chen, & H. Lu, “MeDSLIP: Medical Dual-Stream Language-Image Pre-training for Fine-grained Alignment”, *arXiv: 2403.10635*, 2025.
- [P2] D. Selvarajah, P. Tripathi, K. Teh, S. Zhou, **M. N. I. Suvon**, S. Tesfaye, J. Wild, D. Bennett, A. Segerdahl, D. Steel, L. Colvin and H. Lu., “Leveraging artificial intelligence for predicting placebo vs treatment response in painful diabetic neuropathy,” *DIABETOLOGIA*, Vol. 67, pp. S444–S444, 2024.

## Honors and Awards

- **Fretwell-Downing Prize** (Best MSc dissertation), Department of Computer Science, University of Sheffield 2022
- MSc Academic Performance: **Distinction Award**, University of Sheffield 2022
- BSc Academic Performance: **Cum Laude (Distinction)**, North South University 2020
- BSc Academic Excellence Scholarship (**25% Tuition Fee Waiver**), North South University 2017–2020

## Selected Research Collaborators

### External:

- David G Kiely, Consultant Respiratory Physician and Director Sheffield Pulmonary Vascular Disease Unit, Sheffield Teaching Hospitals NHS Foundation Trust, UK
- Venet Osmani, Professor of Clinical AI and Machine Learning, Queen Mary University of London, UK
- Lisa Watson, Clinical Research Coordinator, Sheffield Teaching Hospitals NHS Foundation Trust, UK
- Prasun Chandra Tripathi, Assistant Professor, IITRAM, India
- Bishesh Khanal, Founding Director, NAAMII, Nepal
- GM Monsur Habib, Primary Care Respiratory Physician, IPCRG, Bangladesh
- Wei Sang, School of Basic Medical Sciences, Shanxi Medical University, Taiyuan, China
- Peizhen Bai, Senior Machine Learning Scientist, AstraZeneca, UK
- Sina Tabakhi, Postdoctoral Researcher in Bioinformatics, Institute of Cancer Research, London, UK

### Internal at the University of Sheffield:

- School of Medicine and Population Health:
  - Andy Swift, Professor of Cardiothoracic Radiology
  - Samer Alabed, Senior Clinical Research Fellow
  - Harry Hill, Senior Research Fellow
- School of Computer Science:
  - Haiping Lu, Professor of Machine Learning
  - Shuo Zhou, Lecturer in Machine Learning
  - Chen Chen, Lecturer in Computer Vision
  - Donghwan Shin, Senior Lecturer in Software Testing
  - Xianyuan Liu, Senior AI Research Engineer

## Selected Talks

- “Model-Based PD-1 High/Low Gating in CD8 T Cells for Tumour Microenvironment Profiling from Single-Cell Surface Proteomics”, AI, Engineering Biology and Beyond, the University of Bristol, Bristol, January 2026.
- “Multimodal Latent Fusion of ECG Leads for Early Assessment of Pulmonary Hypertension”, Third Workshop on Multimodal AI, Barbican Centre, London, September 2025.
- “Multimodal Variational Autoencoder for Low-cost Cardiac Hemodynamics Instability Detection”, Second Workshop on Multimodal AI, the University of Sheffield, Sheffield, June 2024.

- “Multimodal Cardiothoracic Disease Prediction”, First Multimodal AI Research Sprint, the Alan Turing Institute, London, Nov. 2023.

## Open-source Software

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- Core contributor, and maintainer of [PyKale](#), the only multimodal training library in PyTorch Landscape (450+ GitHub stars).

## Professional Services

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- **Grant Writing and Funding Applications:**
  - NIHR Programme Development Grant: Co-applicant, PIs: Professor Andy Swift & Professor David G Kiely (Submitted)
  - UKRI Generative AI Hub: Dataset Creation Challenge: Co-applicant, PI: Dr Chen Chen (Submitted)
- **Selected Journal Reviewer:**
  - Nucleic Acids Research (IF: 13.1)
  - IEEE Transactions on Cognitive and Developmental Systems (IF: 4.9)
  - Bioinformatics (IF: 5.4)
  - IEEE Access (IF: 3.6)
  - Pest Management Science (IF: 3.8)
- **Selected Conference & Workshop Reviewer:**
  - International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI, 2026)
  - International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI, 2025)
  - MICCAI Workshop on Advancing Data Solutions in Medical Imaging AI (ADSMI @ MICCAI 2024)
  - 6th International Conference on Recent Trends in Image Processing & Pattern Recognition (2023)
- **Technical Support** at
  - Second Multimodal AI Community Forum, online, Mar. 2026
  - Third Workshop on Multimodal AI, London, Sept. 2025
  - AI UK Fringe Event: The First Multimodal AI Community Forum, online, Mar. 2024
  - The Second Workshop on Multimodal AI, Sheffield, June 2024
  - The First Multimodal AI Research Sprint, London, Nov. 2023
- **Organizer** at
  - Mini-Hackathon on Multimodal AI, Third Workshop on Multimodal AI, London, Sept. 2025
  - EMBC Workshop on “Open Biomedical Multimodal AI Research: From Pixels to Molecules”, Copenhagen, Denmark, July 2025
- **Executive Chair** at MICCAI Workshop on Advancing Data Solutions in Medical Imaging AI, Morocco, Oct. 2024.
- **Creator & Core maintainer** of [UK Open Multimodal AI Network](#) website (2023–2026)
- **Core Contributor** of [Open Biomedical Multimodal AI](#) tutorial, 2025
- Contributed to [An Introduction to Transparent Machine Learning](#), an online learning course in Responsible AI supported by the Alan Turing Institute, 2022.